Supplementary Material: Case-Disjoint Benchmark of Surrogate Models for Flow and Thermal Fields in Supercritical CO_2 Natural-Circulation Loops

This supplementary document provides:

- **Section S1:** Detailed loop geometry, pipe/wall/jacket radial dimensions, cross-sectional plane sampling counts, plane-weighted mean CO_2 density tests, and CoolProp $(\partial p/\partial T)_\rho$ values across operating groups.
- **Figure S1:** Spatial field reconstructions and absolute error maps for the median (rank 21/42) and worst (rank 42/42) held-out cases by out-of-fold case-nMAE rank for density (ρ, `DeepONet`), temperature (T, `PINN continuity`), and vertical velocity (v, `Domain-informed FiLM-MLP`).
- **Tables S1–S7:** Complete three-seed (2026, 2027, 2028) case-disjoint and within-group mesh-aligned field metrics across all evaluated models for density (ρ), temperature (T), turbulence eddy dissipation (ε), turbulence kinetic energy (k), and velocity components (u, v, w).
- **Table S8:** Complete three-seed gauge-pressure (p_{gauge}) decomposition into case-level mean pressure error and centered spatial residual error.
- **Table S9:** Complete three-seed leave-one-configuration/geometry-group-out extrapolation metrics across the four structural groups (`HHHC/small`, `HHVC/small`, `VHVC/small`, and `VHVC/large`; 16 models × 8 fields = 128 model-field pairs per group).
- **Table S10:** Exploratory nonparametric Friedman repeated-measures ranks and paired Wilcoxon–Holm tests across the 42 cases.

S1. Detailed Conjugate Loop Geometry, Plane Sampling, and Thermodynamic Derivative Summary

- **Small-scale loops (`HHHC`, 20 cases; `HHVC`, 5 cases; `VHVC`, 4 cases):** Centerline rectangular loop width $L_x = 0.8936$ m ($x_{\text{cl}} \in [0, 0.8936]$ m) and height $L_y = 1.0000$ m ($y_{\text{cl}} \in [0, 1.0000]$ m). The primary sCO_2 fluid flows inside a circular pipe of inner diameter $D_i = 15.0$ mm ($R_i = 7.50$ mm, $z \in [-7.50, +7.50]$ mm) surrounded by a 2.83 mm solid stainless-steel wall ($7.50 \le r \le 10.33$ mm, $\rho = 7969.4$ kg m^{-3}) and a 0.660 m concentric cooling-water jacket annulus ($10.33 \le r \le 12.96$ mm, outer jacket wall $12.96 \le r \le 15.65$ mm). In `HHHC`, the heater is on the bottom leg ($y_{\text{cl}} = 0$) and the cooler jacket spans $x \in [0.1168, 0.7768]$ m on the top leg ($y_{\text{cl}} = 1.000$ m). In `HHVC` and small-loop `VHVC`, the cooler jacket is on the right vertical leg ($x_{\text{cl}} = 0.8936$ m), spanning $y \in [0.2232, 0.8832]$ m and $y \in [0.2792, 0.9392]$ m, respectively, while the heater is on the bottom leg (`HHVC`) or lower left vertical leg (`VHVC`). Each small-loop case is exported on 97 cross-sectional planes (46 constant-x and 51 constant-y planes at 0.02 m nominal spacing), yielding 89,647–97,264 primary sCO_2 nodes per case ($\rho \le 950$ kg m^{-3}) after removing solid wall rows and 7,236–7,430 outer cooling-water jacket nodes ($\rho \in [971.4, 1000.7]$ kg m^{-3}) and deduplicating intersecting plane coordinates.
- **Large-scale loop (`VHVC`, 13 cases):** Centerline width $L_x = 3.0100$ m ($x_{\text{cl}} \in [0, 3.0100]$ m) and height $L_y = 4.0800$ m ($y_{\text{cl}} \in [0, 4.0800]$ m), with inner sCO_2 pipe diameter $D_i = 13.84$ mm ($R_i = 6.92$ mm, $z \in [-6.92, +6.92]$ mm), 3.58 mm solid stainless-steel wall ($6.92 \le r \le 10.50$ mm, $\rho = 7971.8$ kg m^{-3}), and a 1.130 m vertical cooler jacket on the right leg ($x_{\text{cl}} = 3.0100$ m, $y \in [2.220, 3.350]$ m, cooling-water annulus $10.50 \le r \le 38.93$ mm, outer jacket wall $38.93 \le r \le 45.80$ mm). Each large-loop case is sampled on 142 unique cross-sectional planes (61 constant-x and 81 constant-y planes at 0.05 m nominal spacing; raw exports contain 284 plane blocks due to a duplicated plane pass removed during coordinate deduplication), yielding 207,586 primary sCO_2 nodes per case (and 82,185 outer cooling-water jacket nodes).

Table 1: *

Summary of plane-weighted mean CO_2 density $\bar{\rho}_{\mathrm{plane},c}$, node-mean density $\bar{\rho}_c$, loop-mean temperature $\bar{T}_c$, case-mean gauge pressure $\bar{p}_c$, and CoolProp $(\partial p/\partial T)_\rho$ evaluated at $(\bar{\rho}_c, \bar{T}_c)$ across operating subgroups.

Group	$(p_{\mathrm{op}}, T_{\mathrm{in}})$	Cases	Planes	$\bar{\rho}_{\mathrm{plane},c}$ (kg m^{-3})	SD($\bar{\rho}_{\mathrm{plane},c}$)	$\bar{T}_c$ (°C)	$\bar{p}_c$ (MPa)	CoolProp $(\partial p/\partial T)_\rho$ (MPa K^{-1})
`HHHC/small`	80 bar, 305 K	3	97	582.0–612.5	16.06	34.43–35.85	0.360–0.464	0.235–0.255
`HHHC/small`	90 bar, 300 K	3	97	737.7–756.1	9.23	30.89–33.24	0.281–0.553	0.393–0.417
`HHHC/small`	90 bar, 305 K	8	97	671.6–704.0	9.79	33.42–36.14	0.223–0.504	0.315–0.349
`HHHC/small`	90 bar, 315 K	3	97	376.2–389.4	6.62	45.28–47.52	0.365–0.520	0.133–0.139
`HHHC/small`	100 bar, 305 K	3	97	709.1–723.2	7.09	35.12–37.38	0.319–0.631	0.362–0.377
`HHVC/small`	90 bar, 305 K	5	97	681.2–717.8	16.66	31.92–39.36	0.006–1.763	0.332–0.365
`VHVC/small`	90 bar, 305 K	4	97	696.8–712.1	6.62	32.63–34.59	0.131–0.513	0.342–0.358
`VHVC/large`	85 bar, 283.5 K	13	142	161.5–860.8	270.83	16.48–92.22	0.001–0.009	0.039–0.596

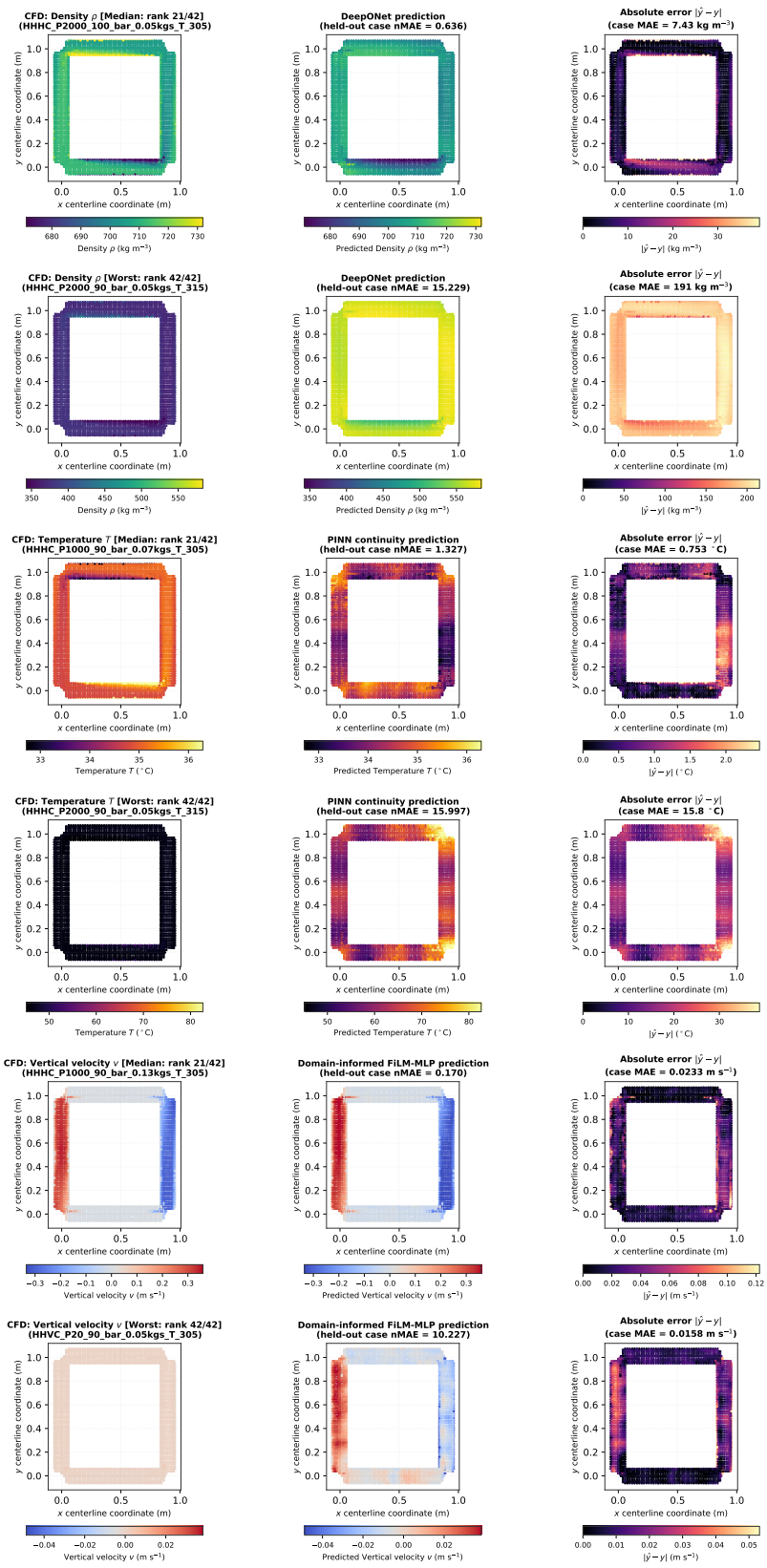


Figure 1: *

Figure S1: Out-of-fold spatial reconstructions and absolute error maps for the median (rank 21/42) and worst (rank 42/42) held-out cases selected by case-nMAE rank (seed 2026) for density ρ (`DeepONet`), temperature T (`PINN continuity`), and vertical velocity v (`Domain-informed FiLM-MLP`).

Table 2: Table S1: Complete three-seed (2026, 2027, 2028) out-of-fold benchmark metrics for Density (ρ, kg m^{-3}) across all 42 primary sCO_2 cases. Entries report three-seed mean ± SD of median case nMAE, mean interquartile range (IQR), mean 95% case-bootstrap CI, case-macro MAE, case-macro RMSE, and pooled R^2.

Model	Protocol	Median nMAE (±SD)	nMAE IQR	95% Bootstrap CI	Macro MAE	Macro RMSE	Pooled R^2
DeepONet	Case-disjoint 5-fold	0.837 ± 0.16	[0.473, 2.24]	[0.611, 1.71]	35.1	37.1	0.862
POD-RBF	Within-group aligned	0.966 ± 0.28	[0.295, 2.06]	[0.543, 1.46]	34.4	35.2	0.883
Domain-informed Concat-MLP	Case-disjoint 5-fold	1.01 ± 0.069	[0.583, 2.3]	[0.719, 1.5]	26.3	29.1	0.936
Domain-informed FiLM-MLP	Case-disjoint 5-fold	1.04 ± 0.26	[0.447, 2.49]	[0.632, 1.7]	31.3	33.2	0.913
PINN data-only	Case-disjoint 5-fold	1.16 ± 0.098	[0.617, 2]	[0.858, 1.43]	30.6	33.8	0.914
PINN continuity	Case-disjoint 5-fold	1.17 ± 0.021	[0.658, 2.4]	[0.85, 1.65]	33.6	36.6	0.901
Nearest case	Within-group aligned	1.18 ± 0.15	[0.633, 2.47]	[0.849, 1.76]	45.1	46.5	0.792
GAMS-GNN-style	Case-disjoint 5-fold	1.25 ± 0.14	[0.752, 2.01]	[0.88, 1.65]	31.8	34.8	0.917
Transolver-style	Case-disjoint 5-fold	1.28 ± 0.22	[0.705, 2.14]	[0.869, 1.58]	32.9	35.6	0.903
Latent Neural Operator	Case-disjoint 5-fold	1.32 ± 0.18	[0.651, 2.66]	[0.904, 1.92]	34.1	37.8	0.882
XGBoost	Case-disjoint 5-fold	1.42 ± 0.071	[0.862, 3.66]	[1.08, 2.1]	57.4	59.2	0.676
Extra Trees	Case-disjoint 5-fold	1.46 ± 0.26	[0.643, 2.86]	[0.88, 2]	42.7	44.6	0.849
CatBoost	Case-disjoint 5-fold	1.56 ± 0.061	[0.688, 3.7]	[1.11, 2.36]	56.3	58.1	0.693
POD-GPR	Within-group aligned	1.58 ± 0.17	[0.605, 3.26]	[0.84, 2.72]	46.7	47.6	0.847
Random Forest	Case-disjoint 5-fold	1.63 ± 0.62	[0.865, 3.31]	[1.1, 2.36]	57.9	59.8	0.636
U-FNO3D	Case-disjoint 5-fold	1.77 ± 0.06	[1.09, 3.28]	[1.44, 2.54]	45.6	50.4	0.847
F-FNO	Case-disjoint 5-fold	2.17 ± 0.52	[1.08, 3.42]	[1.56, 2.83]	44.2	49.9	0.856
Ridge	Case-disjoint 5-fold	5.04 ± 0.46	[1.58, 8.72]	[2.57, 7.05]	90.5	93.4	0.586
Mean baseline	Case-disjoint 5-fold	6.54 ± 0.61	[2.7, 12.5]	[3.9, 10.3]	129	132	-0.034

Table 3: Table S2: Complete three-seed (2026, 2027, 2028) out-of-fold benchmark metrics for Temperature (T, $^{\circ}$C) across all 42 primary sCO_2 cases. Entries report three-seed mean ± SD of median case nMAE, mean interquartile range (IQR), mean 95% case-bootstrap CI, case-macro MAE, case-macro RMSE, and pooled R^2.

Model	Protocol	Median nMAE (±SD)	nMAE IQR	95% Bootstrap CI	Macro MAE	Macro RMSE	Pooled R^2
POD-RBF	Within-group aligned	0.974 ± 0.041	[0.291, 2.18]	[0.545, 1.55]	6.11	7.42	0.62
PINN continuity	Case-disjoint 5-fold	1.35 ± 0.044	[0.773, 3.52]	[1.02, 2.34]	5.48	7.32	0.736
POD-GPR	Within-group aligned	1.36 ± 0.054	[0.619, 3.6]	[0.768, 2.12]	7.11	8.67	0.507
DeepONet	Case-disjoint 5-fold	1.42 ± 0.41	[0.782, 4.3]	[1.02, 2.75]	7.06	8.93	0.634
Extra Trees	Case-disjoint 5-fold	1.44 ± 0.17	[0.552, 2.86]	[0.897, 1.87]	5.55	6.93	0.742
Random Forest	Case-disjoint 5-fold	1.45 ± 0.15	[0.907, 2.89]	[1.13, 1.89]	8.28	10.1	0.526
CatBoost	Case-disjoint 5-fold	1.46 ± 0.1	[0.846, 2.98]	[1.1, 2.1]	7.78	9.49	0.593
Domain-informed FiLM-MLP	Case-disjoint 5-fold	1.47 ± 0.25	[0.609, 3.65]	[0.958, 2.38]	5.34	6.79	0.806
Nearest case	Within-group aligned	1.5 ± 0.044	[0.996, 2.54]	[1.17, 2.06]	7.26	8.82	0.621
XGBoost	Case-disjoint 5-fold	1.55 ± 0.024	[0.988, 2.72]	[1.29, 1.99]	7.25	8.75	0.659
PINN data-only	Case-disjoint 5-fold	1.75 ± 0.18	[0.91, 3.31]	[1.23, 2.46]	5.38	7.33	0.76
Domain-informed Concat-MLP	Case-disjoint 5-fold	1.76 ± 0.15	[0.779, 3.78]	[1.15, 2.88]	5.14	6.89	0.792
GAMS-GNN-style	Case-disjoint 5-fold	1.84 ± 0.17	[0.931, 4.16]	[1.23, 3.17]	5.95	7.89	0.706
Transolver-style	Case-disjoint 5-fold	1.93 ± 0.54	[0.871, 4.37]	[1.24, 3.41]	6.38	8.26	0.683
Latent Neural Operator	Case-disjoint 5-fold	2.19 ± 0.62	[0.985, 5.18]	[1.39, 3.84]	6.86	9.06	0.672
U-FNO3D	Case-disjoint 5-fold	3.49 ± 0.065	[1.75, 5.93]	[2.58, 4.84]	7.83	9.85	0.592
F-FNO	Case-disjoint 5-fold	4.13 ± 0.73	[2.05, 7.14]	[2.61, 6.32]	8.62	10.9	0.565
Mean baseline	Case-disjoint 5-fold	6.94 ± 1.6	[3.31, 11.4]	[4.42, 9.34]	14.9	17	-0.0164
Ridge	Case-disjoint 5-fold	8.94 ± 1.3	[2.98, 27.6]	[4.81, 16.8]	18.3	20.7	0.274

Table 4: Table S3: Complete three-seed (2026, 2027, 2028) out-of-fold benchmark metrics for Turbulence eddy dissipation (ε, $m^2\ s^{-3}$) across all 42 primary sCO_2 cases. Entries report three-seed mean ± SD of median case nMAE, mean interquartile range (IQR), mean 95% case-bootstrap CI, case-macro MAE, case-macro RMSE, and pooled R^2.

Model	Protocol	Median nMAE (±SD)	nMAE IQR	95% Bootstrap CI	Macro MAE	Macro RMSE	Pooled R^2
POD-GPR	Within-group aligned	0.15 ± 0.011	[0.0728, 0.297]	[0.106, 0.238]	0.0995	0.33	0.438
POD-RBF	Within-group aligned	0.177 ± 0.018	[0.113, 0.275]	[0.131, 0.236]	0.127	0.369	-0.575
Domain-informed FiLM-MLP	Case-disjoint 5-fold	0.192 ± 0.02	[0.153, 0.259]	[0.169, 0.226]	0.0978	0.315	0.569
Nearest case	Within-group aligned	0.205 ± 0.017	[0.122, 0.297]	[0.144, 0.244]	0.137	0.388	0.431
XGBoost	Case-disjoint 5-fold	0.221 ± 0.007	[0.192, 0.281]	[0.199, 0.258]	0.132	0.391	0.329
Domain-informed Concat-MLP	Case-disjoint 5-fold	0.226 ± 0.0051	[0.193, 0.31]	[0.208, 0.265]	0.121	0.399	0.326
Random Forest	Case-disjoint 5-fold	0.236 ± 0.0073	[0.213, 0.282]	[0.225, 0.26]	0.134	0.391	0.331
PINN data-only	Case-disjoint 5-fold	0.25 ± 0.0093	[0.216, 0.301]	[0.229, 0.273]	0.119	0.376	0.435
CatBoost	Case-disjoint 5-fold	0.258 ± 0.0017	[0.241, 0.294]	[0.247, 0.275]	0.138	0.404	0.288
Latent Neural Operator	Case-disjoint 5-fold	0.261 ± 0.0097	[0.24, 0.315]	[0.248, 0.282]	0.14	0.426	0.176
DeepONet	Case-disjoint 5-fold	0.267 ± 0.02	[0.238, 0.301]	[0.248, 0.28]	0.144	0.436	0.143
PINN continuity	Case-disjoint 5-fold	0.275 ± 0.0062	[0.254, 0.329]	[0.261, 0.304]	0.136	0.422	0.229
Transolver-style	Case-disjoint 5-fold	0.297 ± 0.014	[0.266, 0.332]	[0.278, 0.317]	0.141	0.415	0.28
GAMS-GNN-style	Case-disjoint 5-fold	0.298 ± 0.0024	[0.277, 0.334]	[0.285, 0.316]	0.142	0.404	0.256
Extra Trees	Case-disjoint 5-fold	0.322 ± 0.0044	[0.302, 0.339]	[0.315, 0.33]	0.154	0.438	0.124
U-FNO3D	Case-disjoint 5-fold	0.366 ± 0.0027	[0.348, 0.403]	[0.357, 0.388]	0.177	0.466	0.0686
F-FNO	Case-disjoint 5-fold	0.367 ± 0.0083	[0.347, 0.404]	[0.356, 0.389]	0.174	0.467	0.0351
Ridge	Case-disjoint 5-fold	0.395 ± 0.0011	[0.375, 0.428]	[0.387, 0.413]	0.184	0.483	-0.0257
Mean baseline	Case-disjoint 5-fold	2.93 ± 0.19	[0.668, 5.69]	[1.24, 4.53]	0.3	0.561	-0.00986

Table 5: Table S4: Complete three-seed (2026, 2027, 2028) out-of-fold benchmark metrics for Turbulence kinetic energy (k, $m^2\ s^{-2}$) across all 42 primary sCO_2 cases. Entries report three-seed mean ± SD of median case nMAE, mean interquartile range (IQR), mean 95% case-bootstrap CI, case-macro MAE, case-macro RMSE, and pooled R^2.

Model	Protocol	Median nMAE (±SD)	nMAE IQR	95% Bootstrap CI	Macro MAE	Macro RMSE	Pooled R^2
POD-RBF	Within-group aligned	0.237 ± 0.015	[0.146, 0.401]	[0.199, 0.327]	0.0002	0.000278	-0.0703
Domain-informed FiLM-MLP	Case-disjoint 5-fold	0.242 ± 0.051	[0.154, 0.396]	[0.177, 0.307]	9.47e-05	0.000157	0.898
POD-GPR	Within-group aligned	0.245 ± 0.01	[0.131, 0.423]	[0.171, 0.31]	0.000134	0.000212	0.729
Domain-informed Concat-MLP	Case-disjoint 5-fold	0.273 ± 0.03	[0.213, 0.415]	[0.232, 0.329]	0.000108	0.000185	0.86
XGBoost	Case-disjoint 5-fold	0.309 ± 0.027	[0.232, 0.553]	[0.253, 0.411]	0.000199	0.00028	0.593
Random Forest	Case-disjoint 5-fold	0.312 ± 0.011	[0.263, 0.554]	[0.274, 0.404]	0.000199	0.000281	0.629
PINN data-only	Case-disjoint 5-fold	0.326 ± 0.019	[0.253, 0.438]	[0.271, 0.383]	0.000124	0.000205	0.829
Nearest case	Within-group aligned	0.329 ± 0.037	[0.145, 0.76]	[0.174, 0.495]	0.000225	0.0003	0.601
CatBoost	Case-disjoint 5-fold	0.355 ± 0.009	[0.32, 0.584]	[0.332, 0.426]	0.000199	0.000282	0.628
DeepONet	Case-disjoint 5-fold	0.399 ± 0.033	[0.316, 0.511]	[0.345, 0.447]	0.000173	0.000268	0.659
Latent Neural Operator	Case-disjoint 5-fold	0.403 ± 0.027	[0.314, 0.476]	[0.347, 0.444]	0.000159	0.000248	0.725
Extra Trees	Case-disjoint 5-fold	0.419 ± 0.0028	[0.379, 0.465]	[0.394, 0.448]	0.000174	0.000251	0.7
PINN continuity	Case-disjoint 5-fold	0.431 ± 0.018	[0.387, 0.516]	[0.405, 0.463]	0.000152	0.000242	0.778
Transolver-style	Case-disjoint 5-fold	0.447 ± 0.018	[0.405, 0.55]	[0.421, 0.494]	0.000179	0.000271	0.704
GAMS-GNN-style	Case-disjoint 5-fold	0.461 ± 0.01	[0.407, 0.531]	[0.429, 0.5]	0.000177	0.000267	0.71
F-FNO	Case-disjoint 5-fold	0.52 ± 0.02	[0.475, 0.632]	[0.496, 0.564]	0.000201	0.000298	0.616
U-FNO3D	Case-disjoint 5-fold	0.523 ± 0.013	[0.478, 0.612]	[0.497, 0.565]	0.000202	0.000302	0.559
Ridge	Case-disjoint 5-fold	0.622 ± 0.0053	[0.553, 0.962]	[0.597, 0.867]	0.000311	0.000402	0.212
Mean baseline	Case-disjoint 5-fold	2.58 ± 0.18	[1.36, 4.19]	[1.52, 3.56]	0.000555	0.000625	-0.076

Table 6: Table S5: Complete three-seed (2026, 2027, 2028) out-of-fold benchmark metrics for Horizontal velocity (u, m s^{-1}) across all 42 primary sCO_2 cases. Entries report three-seed mean ± SD of median case nMAE, mean interquartile range (IQR), mean 95% case-bootstrap CI, case-macro MAE, case-macro RMSE, and pooled R^2.

Model	Protocol	Median nMAE (±SD)	nMAE IQR	95% Bootstrap CI	Macro MAE	Macro RMSE	Pooled R^2
Domain-informed FiLM-MLP	Case-disjoint 5-fold	0.162 ± 0.015	[0.0805, 0.479]	[0.111, 0.346]	0.0381	0.0569	0.801
Nearest case	Within-group aligned	0.167 ± 0.033	[0.108, 0.35]	[0.124, 0.239]	0.047	0.0728	0.63
POD-RBF	Within-group aligned	0.167 ± 0.048	[0.0785, 0.39]	[0.0993, 0.288]	0.0407	0.0627	0.749
POD-GPR	Within-group aligned	0.195 ± 0.035	[0.0906, 0.476]	[0.115, 0.334]	0.0461	0.0709	0.705
XGBoost	Case-disjoint 5-fold	0.205 ± 0.051	[0.122, 0.477]	[0.145, 0.331]	0.0435	0.0689	0.715
Random Forest	Case-disjoint 5-fold	0.215 ± 0.014	[0.127, 0.411]	[0.148, 0.352]	0.0413	0.0656	0.76
DeepONet	Case-disjoint 5-fold	0.217 ± 0.046	[0.161, 0.385]	[0.177, 0.311]	0.0426	0.0689	0.771
Latent Neural Operator	Case-disjoint 5-fold	0.221 ± 0.038	[0.154, 0.559]	[0.174, 0.361]	0.0441	0.0685	0.771
Domain-informed Concat-MLP	Case-disjoint 5-fold	0.224 ± 0.057	[0.119, 0.54]	[0.136, 0.426]	0.043	0.0644	0.78
PINN data-only	Case-disjoint 5-fold	0.233 ± 0.037	[0.127, 0.57]	[0.154, 0.433]	0.0447	0.068	0.767
PINN continuity	Case-disjoint 5-fold	0.243 ± 0.017	[0.174, 0.488]	[0.191, 0.389]	0.0468	0.0742	0.75
Transolver-style	Case-disjoint 5-fold	0.265 ± 0.042	[0.206, 0.524]	[0.231, 0.415]	0.049	0.0746	0.758
CatBoost	Case-disjoint 5-fold	0.268 ± 0.0094	[0.177, 0.458]	[0.204, 0.365]	0.0438	0.0657	0.795
GAMS-GNN-style	Case-disjoint 5-fold	0.269 ± 0.034	[0.201, 0.516]	[0.226, 0.423]	0.0496	0.0767	0.739
Extra Trees	Case-disjoint 5-fold	0.291 ± 0.028	[0.149, 0.536]	[0.185, 0.436]	0.052	0.0822	0.685
U-FNO3D	Case-disjoint 5-fold	0.343 ± 0.033	[0.258, 0.622]	[0.282, 0.476]	0.0593	0.0857	0.682
F-FNO	Case-disjoint 5-fold	0.396 ± 0.019	[0.293, 0.685]	[0.328, 0.539]	0.0648	0.091	0.648
Mean baseline	Case-disjoint 5-fold	0.709 ± 0.002	[0.693, 0.728]	[0.699, 0.718]	0.105	0.152	-0.00576
Ridge	Case-disjoint 5-fold	0.748 ± 0.013	[0.703, 0.811]	[0.715, 0.789]	0.11	0.153	0.00274

Table 7: Table S6: Complete three-seed (2026, 2027, 2028) out-of-fold benchmark metrics for Vertical velocity (v, m s^{-1}) across all 42 primary sCO_2 cases. Entries report three-seed mean ± SD of median case nMAE, mean interquartile range (IQR), mean 95% case-bootstrap CI, case-macro MAE, case-macro RMSE, and pooled R^2.

Model	Protocol	Median nMAE (±SD)	nMAE IQR	95% Bootstrap CI	Macro MAE	Macro RMSE	Pooled R^2
Domain-informed FiLM-MLP	Case-disjoint 5-fold	0.175 ± 0.0041	[0.1, 0.382]	[0.122, 0.271]	0.0383	0.0591	0.861
POD-RBF	Within-group aligned	0.186 ± 0.052	[0.0823, 0.393]	[0.124, 0.31]	0.0405	0.0658	0.824
Nearest case	Within-group aligned	0.192 ± 0.038	[0.116, 0.374]	[0.148, 0.318]	0.051	0.0799	0.703
POD-GPR	Within-group aligned	0.215 ± 0.029	[0.0985, 0.47]	[0.146, 0.325]	0.0451	0.0725	0.792
Domain-informed Concat-MLP	Case-disjoint 5-fold	0.23 ± 0.048	[0.127, 0.453]	[0.156, 0.337]	0.0438	0.066	0.845
XGBoost	Case-disjoint 5-fold	0.237 ± 0.048	[0.148, 0.429]	[0.175, 0.338]	0.0487	0.0751	0.764
Latent Neural Operator	Case-disjoint 5-fold	0.249 ± 0.025	[0.188, 0.423]	[0.21, 0.34]	0.0465	0.0713	0.832
Random Forest	Case-disjoint 5-fold	0.251 ± 0.013	[0.157, 0.423]	[0.192, 0.319]	0.0465	0.0725	0.798
PINN data-only	Case-disjoint 5-fold	0.253 ± 0.029	[0.141, 0.451]	[0.167, 0.345]	0.0446	0.0683	0.839
DeepONet	Case-disjoint 5-fold	0.266 ± 0.047	[0.2, 0.407]	[0.232, 0.361]	0.0478	0.075	0.813
CatBoost	Case-disjoint 5-fold	0.293 ± 0.013	[0.223, 0.412]	[0.247, 0.366]	0.0495	0.0749	0.807
PINN continuity	Case-disjoint 5-fold	0.322 ± 0.0079	[0.209, 0.467]	[0.272, 0.411]	0.0517	0.0826	0.799
Extra Trees	Case-disjoint 5-fold	0.341 ± 0.016	[0.232, 0.551]	[0.262, 0.411]	0.056	0.0879	0.753
Transolver-style	Case-disjoint 5-fold	0.351 ± 0.019	[0.277, 0.507]	[0.315, 0.455]	0.0567	0.0858	0.783
GAMS-GNN-style	Case-disjoint 5-fold	0.353 ± 0.022	[0.26, 0.482]	[0.306, 0.434]	0.0565	0.0865	0.78
U-FNO3D	Case-disjoint 5-fold	0.419 ± 0.027	[0.323, 0.555]	[0.36, 0.478]	0.0667	0.0961	0.716
F-FNO	Case-disjoint 5-fold	0.452 ± 0.05	[0.344, 0.577]	[0.385, 0.54]	0.0698	0.0981	0.704
Ridge	Case-disjoint 5-fold	0.772 ± 0.0049	[0.682, 0.847]	[0.733, 0.813]	0.12	0.159	0.112
Mean baseline	Case-disjoint 5-fold	0.783 ± 0.0042	[0.699, 0.814]	[0.716, 0.809]	0.118	0.165	-0.017

Table 8: Table S7: Complete three-seed (2026, 2027, 2028) out-of-fold benchmark metrics for Transverse velocity (w, m s^{-1}) across all 42 primary sCO_2 cases. Entries report three-seed mean ± SD of median case nMAE, mean interquartile range (IQR), mean 95% case-bootstrap CI, case-macro MAE, case-macro RMSE, and pooled R^2.

Model	Protocol	Median nMAE (±SD)	nMAE IQR	95% Bootstrap CI	Macro MAE	Macro RMSE	Pooled R^2
Nearest case	Within-group aligned	0.185 ± 0.0091	[0.111, 0.28]	[0.16, 0.225]	0.00108	0.00293	0.551
POD-GPR	Within-group aligned	0.203 ± 0.012	[0.0928, 0.294]	[0.164, 0.254]	0.00101	0.00256	0.665
POD-RBF	Within-group aligned	0.207 ± 0.0079	[0.102, 0.302]	[0.151, 0.256]	0.00101	0.00255	0.65
Domain-informed FiLM-MLP	Case-disjoint 5-fold	0.272 ± 0.007	[0.205, 0.366]	[0.238, 0.338]	0.00143	0.00346	0.418
XGBoost	Case-disjoint 5-fold	0.318 ± 0.0045	[0.206, 0.36]	[0.301, 0.329]	0.00144	0.00356	0.43
Domain-informed Concat-MLP	Case-disjoint 5-fold	0.318 ± 0.018	[0.267, 0.405]	[0.288, 0.362]	0.00156	0.00385	0.311
PINN data-only	Case-disjoint 5-fold	0.326 ± 0.0026	[0.271, 0.397]	[0.296, 0.368]	0.00159	0.00368	0.37
Random Forest	Case-disjoint 5-fold	0.33 ± 0.012	[0.227, 0.356]	[0.299, 0.343]	0.00149	0.00409	0.233
Extra Trees	Case-disjoint 5-fold	0.345 ± 0.0028	[0.231, 0.368]	[0.303, 0.356]	0.00152	0.00437	0.128
Latent Neural Operator	Case-disjoint 5-fold	0.355 ± 0.01	[0.288, 0.41]	[0.334, 0.379]	0.0017	0.004	0.273
CatBoost	Case-disjoint 5-fold	0.362 ± 0.0036	[0.223, 0.381]	[0.348, 0.371]	0.00155	0.00404	0.257
Mean baseline	Case-disjoint 5-fold	0.363 ± 0.0002	[0.238, 0.383]	[0.307, 0.376]	0.00158	0.0047	-1.91e-05
Ridge	Case-disjoint 5-fold	0.364 ± 0.00073	[0.24, 0.387]	[0.325, 0.379]	0.00159	0.0047	9.99e-05
PINN continuity	Case-disjoint 5-fold	0.364 ± 0.00098	[0.242, 0.384]	[0.314, 0.378]	0.00158	0.00469	0.00378
DeepONet	Case-disjoint 5-fold	0.365 ± 0.0053	[0.265, 0.387]	[0.337, 0.38]	0.00166	0.00423	0.168
GAMS-GNN-style	Case-disjoint 5-fold	0.376 ± 0.0015	[0.266, 0.394]	[0.359, 0.386]	0.00165	0.00469	0.00143
Transolver-style	Case-disjoint 5-fold	0.378 ± 0.0013	[0.262, 0.396]	[0.359, 0.39]	0.00166	0.00469	0.00342
U-FNO3D	Case-disjoint 5-fold	0.392 ± 0.0033	[0.307, 0.423]	[0.376, 0.41]	0.00174	0.00473	-0.00833
F-FNO	Case-disjoint 5-fold	0.4 ± 0.0041	[0.32, 0.435]	[0.383, 0.415]	0.0018	0.00474	-0.0152

Table 9: Table S8: Complete three-seed gauge-pressure (p_{gauge}) decomposition metrics across all 42 cases (19 competing models plus `Lattice floor (oracle)`). Columns report total node-level case-macro MAE and RMSE (kPa), case-mean offset MAE and RMSE (kPa), centered spatial residual MAE and 42-case root-mean-square residual RMSE (Pa), median centered residual nRMSE (with a 1 kPa denominator floor), and pooled gauge-pressure R^2.

Model	Protocol	Macro MAE (kPa)	Macro RMSE (kPa)	Level MAE (kPa)	Level RMSE (kPa)	Resid. MAE (Pa)	Resid. RMSE (Pa)	Resid. nRMSE	Pool
Lattice floor (oracle)	Oracle reference	0.00936 ± 3.6e-07	0.0141	1.59e-05	2.6e-05	9.36	26.8	0.00185	
Extra Trees	Case-disjoint 5-fold	83.2 ± 1.4	83.7	82.9	160	4.55e+03	1.01e+04	2.91	
POD-GPR	Within-group aligned	86 ± 7.8	86	86	131	201	457	0.0457	
POD-RBF	Within-group aligned	90.1 ± 7.5	90.2	90.1	244	201	471	0.0483	
Random Forest	Case-disjoint 5-fold	91.5 ± 13	92.5	91.1	174	5.26e+03	9.44e+03	3.61	
Domain-informed Concat-MLP	Case-disjoint 5-fold	95.4 ± 13	102	90.3	172	2.4e+04	4.2e+04	18.3	
Domain-informed FiLM-MLP	Case-disjoint 5-fold	95.7 ± 19	99.4	93.3	193	1.65e+04	4.41e+04	10.7	
Latent Neural Operator	Case-disjoint 5-fold	97.2 ± 15	105	90	172	2.95e+04	6.05e+04	19.8	
PINN data-only	Case-disjoint 5-fold	102 ± 4.9	109	95.4	191	2.7e+04	4.51e+04	22.2	
CatBoost	Case-disjoint 5-fold	104 ± 3.8	104	104	200	1.56e+03	3.39e+03	0.954	
Nearest case	Within-group aligned	108 ± 6.3	108	108	175	271	667	0.0122	
DeepONet	Case-disjoint 5-fold	111 ± 7.6	111	111	201	5.58e+03	8.55e+03	4.53	
XGBoost	Case-disjoint 5-fold	121 ± 3.4	121	121	257	427	1.01e+03	0.151	
PINN continuity	Case-disjoint 5-fold	131 ± 25	136	127	276	2.39e+04	4.8e+04	13.4	
GAMS-GNN-style	Case-disjoint 5-fold	141 ± 10	144	138	319	1.3e+04	1.97e+04	9.76	
Transolver-style	Case-disjoint 5-fold	149 ± 15	151	147	327	9.17e+03	1.34e+04	8.15	
U-FNO3D	Case-disjoint 5-fold	167 ± 7.7	173	162	330	2.45e+04	3.47e+04	21.5	
F-FNO	Case-disjoint 5-fold	174 ± 13	181	164	334	2.98e+04	4.34e+04	24.3	
Ridge	Case-disjoint 5-fold	175 ± 15	175	175	320	1.62e+03	2.56e+03	1.07	
Mean baseline	Case-disjoint 5-fold	236 ± 3.9	236	236	343	1.35e+03	2.89e+03	0.202	

Table 10: Table S9: Complete three-seed leave-one-configuration/geometry-group-out extrapolation metrics across the four structural groups. Entries report three-seed mean of median case nMAE for the seven spatially varying fields, case-level gauge-pressure MAE (kPa), and centered pressure residual nRMSE (1 kPa floor). Mesh-aligned transfer methods (POD-RBF, POD-GPR, Nearest case) are not applicable across disjoint loop geometries.

Held-out group	Model	ρ nMAE	T nMAE	ε nMAE	k nMAE	u nMAE	v nMAE	w nMAE	p Level MAE (kPa)	p Resid. nRMSE
HHHC / small (n = 20)	CatBoost	8.38	13.2	4.63	6.92	1.03	0.953	0.48	270	10.7
HHHC / small (n = 20)	DeepONet	9.66	68.8	0.409	0.974	0.971	0.991	0.393	940	20.5
HHHC / small (n = 20)	Domain-informed Concat-MLP	12.9	130	1.29	1.31	0.866	0.876	0.891	2.04e+03	67.9
HHHC / small (n = 20)	Domain-informed FiLM-MLP	12.1	113	6.82	1.3	0.973	1.05	0.514	1.91e+03	122
HHHC / small (n = 20)	Extra Trees	7	11.6	0.409	1.03	0.938	1.01	0.391	805	1.34
HHHC / small (n = 20)	F-FNO	9.4	45.1	0.402	0.652	0.794	0.934	0.393	138	18.7
HHHC / small (n = 20)	GAMS-GNN-style	4.6	16.2	0.476	0.827	0.721	0.79	0.438	123	21.4
HHHC / small (n = 20)	Latent Neural Operator	7.41	27.5	0.753	0.946	0.997	1.08	0.395	1.07e+03	115
HHHC / small (n = 20)	Mean baseline	5.11	9.57	5.85	4.63	0.748	0.805	0.377	141	0.196
HHHC / small (n = 20)	PINN continuity	5.48	44.2	1.01	1.58	0.688	0.734	0.386	1.18e+03	12.6
HHHC / small (n = 20)	PINN data-only	5.46	31.5	0.817	1.2	0.817	0.723	0.773	1.1e+03	46.9
HHHC / small (n = 20)	Random Forest	8.87	13.1	5.75	5.47	0.896	1.05	0.544	525	2.12
HHHC / small (n = 20)	Ridge	21.8	128	0.405	1.06	0.974	1.12	0.379	195	2.22
HHHC / small (n = 20)	Transolver-style	6.12	32.3	2.19	1.12	0.751	0.96	0.451	123	37.5
HHHC / small (n = 20)	U-FNO3D	9.92	32.7	0.438	1.04	0.84	0.917	0.407	131	45.4
HHHC / small (n = 20)	XGBoost	11.9	18.1	7.51	10.7	0.933	1.19	0.47	247	5.07
HHVC / small ($n = 5$)	CatBoost	1.51	3.23	0.298	0.505	0.791	0.666	0.508	564	0.773
HHVC / small ($n = 5$)	DeepONet	6.11	7.94	0.391	0.698	2.26	1.49	0.785	640	8.96
HHVC / small ($n = 5$)	Domain-informed Concat-MLP	17.1	34.7	29.5	1.59	1.94	1.53	1.77	1.03e+03	98.5
HHVC / small ($n = 5$)	Domain-informed FiLM-MLP	12.1	13.9	0.896	0.876	1.75	1.14	0.982	795	16.2
HHVC / small ($n = 5$)	Extra Trees	1.17	3.34	0.327	0.618	2.02	1.21	0.416	582	3.73
HHVC / small ($n = 5$)	F-FNO	11.3	47.3	0.386	1.53	2.24	1.27	0.927	615	24.5
HHVC / small ($n = 5$)	GAMS-GNN-style	12.3	15.6	4.06	2.24	2.64	1.63	0.618	607	29.8
HHVC / small ($n = 5$)	Latent Neural Operator	11.4	20.6	0.441	0.756	2.11	1.53	0.833	671	27.6
HHVC / small ($n = 5$)	Mean baseline	13.3	21.8	15.7	9.95	0.998	0.95	0.375	608	0.186
HHVC / small ($n = 5$)	PINN continuity	4.94	19.2	0.491	0.659	1.79	1.11	0.384	891	46.7
HHVC / small ($n = 5$)	PINN data-only	10.8	13.3	20.8	1.69	1.23	0.875	0.907	726	79.4
HHVC / small ($n = 5$)	Random Forest	1.51	3.44	0.281	0.445	1.69	0.985	0.43	599	3.92
HHVC / small ($n = 5$)	Ridge	20.7	82.9	0.283	0.436	0.829	0.702	0.377	598	1.42
HHVC / small ($n = 5$)	Transolver-style	20.1	10.2	1.97	1.51	2.64	1.69	0.64	611	29.2
HHVC / small ($n = 5$)	U-FNO3D	13.2	10.3	0.432	1.12	1.86	1.26	0.662	603	22.1
HHVC / small ($n = 5$)	XGBoost	1.39	3.05	0.31	0.524	3.11	1.41	0.763	557	0.218
VHVC / small ($n = 4$)	CatBoost	1.62	2.43	0.521	1.73	2.6	1.01	0.895	205	2.95
VHVC / small ($n = 4$)	DeepONet	3.5	7.18	0.365	0.85	1.19	0.784	0.508	165	15.3
VHVC / small ($n = 4$)	Domain-informed Concat-MLP	3.24	11.6	2.51e+03	14	4.07	2.53	1.4	165	115
VHVC / small ($n = 4$)	Domain-informed FiLM-MLP	7.42	16.4	14.4	1.09	4.61	1.22	0.812	350	37.1
VHVC / small ($n = 4$)	Extra Trees	2.25	3.74	0.365	0.988	2.56	0.89	0.45	137	3.92
VHVC / small ($n = 4$)	F-FNO	6.81	16.2	0.424	1.39	4.81	3.23	0.873	171	59.4
VHVC / small ($n = 4$)	GAMS-GNN-style	5.29	10.4	1.12	2.46	2.77	2.06	0.523	223	39.5

Held-out group	Model	ρ nMAE	T nMAE	ε nMAE	k nMAE	u nMAE	v nMAE	w nMAE	p Level MAE (kPa)	p Resid. nRMSE
VHVC / small ($n = 4$)	Latent Neural Operator	4.73	7.4	0.528	1.01	1.92	1.08	0.796	246	41.1
VHVC / small ($n = 4$)	Mean baseline	4.51	8.51	9.04	6.79	1.88	1.08	0.39	128	0.185
VHVC / small ($n = 4$)	PINN continuity	6.57	12.5	18.9	2.12	1.56	0.765	0.395	271	87
VHVC / small ($n = 4$)	PINN data-only	2.72	5.2	134	10.5	4.34	1.42	1.28	225	99.2
VHVC / small ($n = 4$)	Random Forest	2.32	3.75	0.383	0.861	1.36	0.699	1.01	201	1.9
VHVC / small ($n = 4$)	Ridge	8.68	27.5	0.437	1.23	1.18	0.72	0.421	144	1.38
VHVC / small ($n = 4$)	Transolver-style	3.85	9.48	1.39	1.41	5.76	1.72	0.519	162	34.2
VHVC / small ($n = 4$)	U-FNO3D	5.5	4.88	0.426	1.6	4.73	1.91	0.879	136	49.1
VHVC / small ($n = 4$)	XGBoost	0.98	1.61	0.282	0.468	1.12	0.686	0.568	386	1.29
VHVC / large ($n = 13$)	CatBoost	5.13	1.8	3.17	1.98	0.97	1.02	0.806	449	1.68
VHVC / large ($n = 13$)	DeepONet	5.9	2.5	5.91e+07	8.81e+07	0.643	0.824	0.915	888	8.41
VHVC / large ($n = 13$)	Domain-informed Concat-MLP	1.77e+04	1.42e+04	1.62e+09	1.36e+11	1.36e+04	4.05e+03	2.6e+04	6.93e+06	12.1
VHVC / large ($n = 13$)	Domain-informed FiLM-MLP	5.29	2.5	25	1.87	0.787	0.857	0.986	646	28.6
VHVC / large ($n = 13$)	Extra Trees	4.36	1.65	0.437	1.98	0.656	0.766	0.261	728	1.3
VHVC / large ($n = 13$)	F-FNO	1.81e+04	8.27e+03	1.62e+09	1.36e+11	1.9e+03	1.64e+04	2.02e+04	1.44e+06	2.17
VHVC / large ($n = 13$)	GAMS-GNN-style	5.25	1.94	0.448	1.8	0.711	0.871	0.333	389	1.09
VHVC / large ($n = 13$)	Latent Neural Operator	4.87	1.87	0.555	1.82	0.919	0.964	0.729	248	20.7
VHVC / large ($n = 13$)	Mean baseline	5.57	1.69	0.431	1.88	0.644	0.836	0.225	431	1
VHVC / large ($n = 13$)	PINN continuity	5.36	3.18	4.46	2.47	0.732	0.909	0.242	1.38e+03	1
VHVC / large ($n = 13$)	PINN data-only	6.01	3.14	1.13	1.16	0.791	0.849	2.84	842	1
VHVC / large ($n = 13$)	Random Forest	3.85	1.98	0.424	1.89	0.967	1.05	0.442	414	3.5
VHVC / large ($n = 13$)	Ridge	2.26e+04	1.28e+04	1.62e+09	9.08e+10	193	2.17e+03	1.73e+03	2.04e+07	1
VHVC / large ($n = 13$)	Transolver-style	5.1	1.75	0.436	1.95	0.688	0.876	0.613	366	1
VHVC / large ($n = 13$)	U-FNO3D	1.42e+04	1.56e+04	1.62e+09	1.36e+11	1.59e+04	7.34e+03	1.79e+04	2.01e+06	3.95e+05
VHVC / large ($n = 13$)	XGBoost	5.85	1.73	7.22	1.83	0.588	0.787	0.83	402	0.988

Table 11: Table S10: Exploratory three-seed nonparametric repeated-measures Friedman ranks, paired case-level median differences $\Delta = \mathrm{Median}_c(e_{\mathrm{comp},c} - e_{\mathrm{FiLM},c})$ with 95% bootstrap confidence intervals, and paired two-sided Wilcoxon signed-rank tests (with Holm step-down multiplicity adjustment against `Domain-informed FiLM-MLP`) across the $N = 42$ held-out cases for all 19 evaluated surrogate models.

Target field	Model	Mean rank	Median Δ vs. FiLM [95% CI]	Friedman χ^2_F	Friedman p	Wilcoxon raw p	Holm p
Density (ρ, kg m^{-3})	Domain-informed Concat-MLP	6.19	−0.060 [−0.169, +0.042]	219	< 0.001	0.171	1
Density (ρ, kg m^{-3})	Domain-informed FiLM-MLP	6.74	– (FiLM ref.)	219	< 0.001	– (FiLM ref.)	– (FiLM ref.)
Density (ρ, kg m^{-3})	POD-RBF	6.74	−0.123 [−0.419, +0.259]	219	< 0.001	0.528	1
Density (ρ, kg m^{-3})	PINN data-only	7.14	+0.049 [−0.093, +0.178]	219	< 0.001	0.683	1
Density (ρ, kg m^{-3})	DeepONet	7.36	+0.076 [−0.125, +0.206]	219	< 0.001	0.536	1
Density (ρ, kg m^{-3})	Nearest case	8.38	+0.123 [−0.298, +0.732]	219	< 0.001	0.465	1
Density (ρ, kg m^{-3})	PINN continuity	8.43	+0.202 [−0.039, +0.307]	219	< 0.001	0.11	0.991
Density (ρ, kg m^{-3})	Latent Neural Operator	8.67	+0.110 [−0.061, +0.160]	219	< 0.001	0.56	1
Density (ρ, kg m^{-3})	Transolver-style	8.71	+0.161 [+0.064, +0.335]	219	< 0.001	0.214	1
Density (ρ, kg m^{-3})	GAMS-GNN-style	8.81	+0.158 [−0.051, +0.243]	219	< 0.001	0.427	1
Density (ρ, kg m^{-3})	Extra Trees	9.5	+0.177 [−0.004, +0.792]	219	< 0.001	0.017	0.17
Density (ρ, kg m^{-3})	POD-GPR	10.6	+0.832 [+0.145, +1.114]	219	< 0.001	< 0.001	0.00215
Density (ρ, kg m^{-3})	CatBoost	11.1	+0.559 [+0.273, +1.270]	219	< 0.001	< 0.001	0.00741
Density (ρ, kg m^{-3})	Random Forest	11.4	+0.643 [+0.211, +1.526]	219	< 0.001	< 0.001	0.00974
Density (ρ, kg m^{-3})	XGBoost	12.3	+0.918 [+0.406, +1.423]	219	< 0.001	< 0.001	0.00224
Density (ρ, kg m^{-3})	U-FNO3D	12.6	+0.710 [+0.389, +1.279]	219	< 0.001	< 0.001	< 0.001
Density (ρ, kg m^{-3})	F-FNO	12.9	+0.668 [+0.330, +1.288]	219	< 0.001	< 0.001	< 0.001
Density (ρ, kg m^{-3})	Ridge	15.1	+2.735 [+1.352, +4.801]	219	< 0.001	< 0.001	< 0.001
Density (ρ, kg m^{-3})	Mean baseline	17.3	+5.054 [+3.069, +9.137]	219	< 0.001	< 0.001	< 0.001
Temperature (T, °C)	POD-RBF	5.88	−0.708 [−1.382, −0.048]	266	< 0.001	0.00363	0.0471
Temperature (T, °C)	PINN continuity	7.05	−0.013 [−0.162, +0.102]	266	< 0.001	0.528	1
Temperature (T, °C)	Extra Trees	7.12	−0.142 [−0.764, +0.110]	266	< 0.001	0.297	1
Temperature (T, °C)	Domain-informed FiLM-MLP	7.31	– (FiLM ref.)	266	< 0.001	– (FiLM ref.)	– (FiLM ref.)
Temperature (T, °C)	Domain-informed Concat-MLP	7.81	+0.094 [+0.002, +0.288]	266	< 0.001	0.263	1
Temperature (T, °C)	PINN data-only	7.93	+0.111 [+0.034, +0.222]	266	< 0.001	0.435	1
Temperature (T, °C)	POD-GPR	8.19	−0.190 [−0.750, +0.299]	266	< 0.001	0.552	1
Temperature (T, °C)	Nearest case	8.38	−0.068 [−0.764, +0.638]	266	< 0.001	0.692	1
Temperature (T, °C)	DeepONet	8.69	+0.151 [−0.222, +0.359]	266	< 0.001	0.243	1
Temperature (T, °C)	CatBoost	9.14	+0.023 [−0.658, +0.451]	266	< 0.001	0.931	1
Temperature (T, °C)	GAMS-GNN-style	9.38	+0.275 [+0.123, +0.556]	266	< 0.001	0.0325	0.358
Temperature (T, °C)	Random Forest	9.52	+0.268 [−0.734, +0.766]	266	< 0.001	0.98	1
Temperature (T, °C)	XGBoost	9.93	+0.324 [−0.516, +0.794]	266	< 0.001	0.457	1
Temperature (T, °C)	Transolver-style	9.95	+0.304 [+0.114, +0.644]	266	< 0.001	0.0285	0.342
Temperature (T, °C)	Latent Neural Operator	10.6	+0.491 [+0.255, +0.821]	266	< 0.001	< 0.001	< 0.001
Temperature (T, °C)	U-FNO3D	14.1	+1.413 [+0.743, +1.934]	266	< 0.001	< 0.001	< 0.001
Temperature (T, °C)	F-FNO	15.1	+2.588 [+1.383, +3.101]	266	< 0.001	< 0.001	< 0.001
Temperature (T, °C)	Mean baseline	16.3	+4.408 [+3.222, +5.885]	266	< 0.001	< 0.001	< 0.001
Temperature (T, °C)	Ridge	17.6	+7.377 [+4.028, +16.473]	266	< 0.001	< 0.001	< 0.001
Turbulence eddy dissipation (ε, m^2 s^{-3})	Domain-informed FiLM-MLP	4.62	– (FiLM ref.)	344	< 0.001	– (FiLM ref.)	– (FiLM ref.)
Turbulence eddy dissipation (ε, m^2 s^{-3})	POD-GPR	6.1	−0.036 [−0.069, −0.001]	344	< 0.001	0.406	1
Turbulence eddy dissipation (ε, m^2 s^{-3})	Nearest case	6.45	+0.004 [−0.043, +0.021]	344	< 0.001	0.766	1
Turbulence eddy dissipation (ε, m^2 s^{-3})	POD-RBF	6.45	−0.023 [−0.062, +0.013]	344	< 0.001	0.99	1
Turbulence eddy dissipation (ε, m^2 s^{-3})	XGBoost	7.19	+0.046 [+0.016, +0.052]	344	< 0.001	0.00254	0.0102
Turbulence eddy dissipation (ε, m^2 s^{-3})	PINN data-only	7.62	+0.057 [+0.032, +0.068]	344	< 0.001	< 0.001	< 0.001

Target field	Model	Mean rank	Median Δ vs. FiLM [95% CI]	Friedman χ^2_F	Friedman p	Wilcoxon raw p	Holm p
Turbulence eddy dissipation (ε, $m^2\ s^{-3}$)	Domain-informed Concat-MLP	7.67	+0.034 [+0.021, +0.070]	344	< 0.001	< 0.001	< 0.001
Turbulence eddy dissipation (ε, $m^2\ s^{-3}$)	Random Forest	7.74	+0.068 [+0.037, +0.078]	344	< 0.001	< 0.001	0.00153
Turbulence eddy dissipation (ε, $m^2\ s^{-3}$)	DeepONet	8.33	+0.073 [+0.056, +0.098]	344	< 0.001	< 0.001	< 0.001
Turbulence eddy dissipation (ε, $m^2\ s^{-3}$)	Latent Neural Operator	8.86	+0.070 [+0.063, +0.082]	344	< 0.001	< 0.001	< 0.001
Turbulence eddy dissipation (ε, $m^2\ s^{-3}$)	CatBoost	9.29	+0.087 [+0.053, +0.098]	344	< 0.001	< 0.001	< 0.001
Turbulence eddy dissipation (ε, $m^2\ s^{-3}$)	PINN continuity	10.7	+0.093 [+0.073, +0.118]	344	< 0.001	< 0.001	< 0.001
Turbulence eddy dissipation (ε, $m^2\ s^{-3}$)	GAMS-GNN-style	11.6	+0.100 [+0.084, +0.117]	344	< 0.001	< 0.001	< 0.001
Turbulence eddy dissipation (ε, $m^2\ s^{-3}$)	Extra Trees	11.7	+0.124 [+0.094, +0.141]	344	< 0.001	< 0.001	< 0.001
Turbulence eddy dissipation (ε, $m^2\ s^{-3}$)	Transolver-style	12	+0.108 [+0.082, +0.123]	344	< 0.001	< 0.001	< 0.001
Turbulence eddy dissipation (ε, $m^2\ s^{-3}$)	F-FNO	14.3	+0.174 [+0.138, +0.194]	344	< 0.001	< 0.001	< 0.001
Turbulence eddy dissipation (ε, $m^2\ s^{-3}$)	U-FNO3D	14.9	+0.179 [+0.147, +0.201]	344	< 0.001	< 0.001	< 0.001
Turbulence eddy dissipation (ε, $m^2\ s^{-3}$)	Ridge	16	+0.196 [+0.168, +0.222]	344	< 0.001	< 0.001	< 0.001
Turbulence eddy dissipation (ε, $m^2\ s^{-3}$)	Mean baseline	18.5	+2.821 [+1.005, +4.553]	344	< 0.001	< 0.001	< 0.001
Turbulence kinetic energy (k, $m^2\ s^{-2}$)	Domain-informed FiLM-MLP	4.62	– (FiLM ref.)	384	< 0.001	– (FiLM ref.)	– (FiLM ref.)
Turbulence kinetic energy (k, $m^2\ s^{-2}$)	POD-GPR	4.98	−0.022 [−0.054, +0.022]	384	< 0.001	0.757	1
Turbulence kinetic energy (k, $m^2\ s^{-2}$)	Domain-informed Concat-MLP	5.1	+0.034 [−0.003, +0.057]	384	< 0.001	0.0475	0.19
Turbulence kinetic energy (k, $m^2\ s^{-2}$)	POD-RBF	5.76	+0.012 [−0.036, +0.058]	384	< 0.001	0.638	1
Turbulence kinetic energy (k, $m^2\ s^{-2}$)	PINN data-only	6.4	+0.071 [+0.020, +0.104]	384	< 0.001	< 0.001	< 0.001
Turbulence kinetic energy (k, $m^2\ s^{-2}$)	Nearest case	7.98	+0.022 [−0.018, +0.147]	384	< 0.001	0.0504	0.19
Turbulence kinetic energy (k, $m^2\ s^{-2}$)	XGBoost	8.31	+0.088 [+0.014, +0.150]	384	< 0.001	< 0.001	0.00309
Turbulence kinetic energy (k, $m^2\ s^{-2}$)	Random Forest	8.52	+0.094 [+0.046, +0.138]	384	< 0.001	< 0.001	< 0.001
Turbulence kinetic energy (k, $m^2\ s^{-2}$)	Latent Neural Operator	8.81	+0.124 [+0.111, +0.149]	384	< 0.001	< 0.001	< 0.001
Turbulence kinetic energy (k, $m^2\ s^{-2}$)	DeepONet	9.29	+0.150 [+0.077, +0.176]	384	< 0.001	< 0.001	< 0.001
Turbulence kinetic energy (k, $m^2\ s^{-2}$)	PINN continuity	10.4	+0.185 [+0.141, +0.219]	384	< 0.001	< 0.001	< 0.001
Turbulence kinetic energy (k, $m^2\ s^{-2}$)	CatBoost	10.4	+0.167 [+0.096, +0.213]	384	< 0.001	< 0.001	< 0.001
Turbulence kinetic energy (k, $m^2\ s^{-2}$)	Extra Trees	10.6	+0.186 [+0.140, +0.230]	384	< 0.001	< 0.001	< 0.001
Turbulence kinetic energy (k, $m^2\ s^{-2}$)	Transolver-style	12.1	+0.223 [+0.173, +0.262]	384	< 0.001	< 0.001	< 0.001
Turbulence kinetic energy (k, $m^2\ s^{-2}$)	GAMS-GNN-style	12.4	+0.230 [+0.180, +0.265]	384	< 0.001	< 0.001	< 0.001
Turbulence kinetic energy (k, $m^2\ s^{-2}$)	U-FNO3D	14.5	+0.289 [+0.247, +0.351]	384	< 0.001	< 0.001	< 0.001
Turbulence kinetic energy (k, $m^2\ s^{-2}$)	F-FNO	14.7	+0.320 [+0.242, +0.375]	384	< 0.001	< 0.001	< 0.001

Target field	Model	Mean rank	Median Δ vs. FiLM [95% CI]	Friedman χ^2_F	Friedman p	Wilcoxon raw p	Holm p
Turbulence kinetic energy (k, $m^2\ s^{-2}$)	Ridge	16.8	+0.422 [+0.355, +0.521]	384	< 0.001	< 0.001	< 0.001
Turbulence kinetic energy (k, $m^2\ s^{-2}$)	Mean baseline	18.4	+2.567 [+1.209, +3.326]	384	< 0.001	< 0.001	< 0.001
Horizontal velocity (u, m s^{-1})	Domain-informed FiLM-MLP	5.83	– (FiLM ref.)	296	< 0.001	– (FiLM ref.)	– (FiLM ref.)
Horizontal velocity (u, m s^{-1})	POD-RBF	6.14	−0.020 [−0.054, +0.007]	296	< 0.001	0.274	1
Horizontal velocity (u, m s^{-1})	POD-GPR	6.98	+0.008 [−0.058, +0.079]	296	< 0.001	0.805	1
Horizontal velocity (u, m s^{-1})	Random Forest	7.36	+0.036 [−0.006, +0.071]	296	< 0.001	0.309	1
Horizontal velocity (u, m s^{-1})	Domain-informed Concat-MLP	7.86	+0.029 [+0.018, +0.055]	296	< 0.001	< 0.001	0.0088
Horizontal velocity (u, m s^{-1})	DeepONet	7.98	+0.047 [+0.005, +0.078]	296	< 0.001	0.0916	0.55
Horizontal velocity (u, m s^{-1})	XGBoost	7.98	+0.045 [−0.009, +0.080]	296	< 0.001	0.131	0.656
Horizontal velocity (u, m s^{-1})	Nearest case	8.07	+0.016 [−0.031, +0.091]	296	< 0.001	0.528	1
Horizontal velocity (u, m s^{-1})	PINN data-only	8.57	+0.046 [+0.020, +0.065]	296	< 0.001	< 0.001	< 0.001
Horizontal velocity (u, m s^{-1})	Latent Neural Operator	8.74	+0.061 [+0.043, +0.083]	296	< 0.001	< 0.001	0.0088
Horizontal velocity (u, m s^{-1})	CatBoost	9	+0.084 [+0.016, +0.096]	296	< 0.001	0.0736	0.515
Horizontal velocity (u, m s^{-1})	Extra Trees	9.33	+0.064 [+0.035, +0.099]	296	< 0.001	0.0127	0.102
Horizontal velocity (u, m s^{-1})	PINN continuity	9.6	+0.066 [+0.047, +0.081]	296	< 0.001	0.00119	0.0107
Horizontal velocity (u, m s^{-1})	GAMS-GNN-style	11.2	+0.093 [+0.063, +0.111]	296	< 0.001	< 0.001	< 0.001
Horizontal velocity (u, m s^{-1})	Transolver-style	11.3	+0.097 [+0.069, +0.116]	296	< 0.001	< 0.001	0.00128
Horizontal velocity (u, m s^{-1})	U-FNO3D	14.3	+0.153 [+0.119, +0.172]	296	< 0.001	< 0.001	< 0.001
Horizontal velocity (u, m s^{-1})	F-FNO	15.7	+0.192 [+0.164, +0.228]	296	< 0.001	< 0.001	< 0.001
Horizontal velocity (u, m s^{-1})	Mean baseline	16.9	+0.562 [+0.544, +0.600]	296	< 0.001	< 0.001	< 0.001
Horizontal velocity (u, m s^{-1})	Ridge	17.2	+0.596 [+0.541, +0.637]	296	< 0.001	< 0.001	< 0.001
Vertical velocity (v, m s^{-1})	Domain-informed FiLM-MLP	5.26	– (FiLM ref.)	333	< 0.001	– (FiLM ref.)	– (FiLM ref.)
Vertical velocity (v, m s^{-1})	POD-RBF	5.5	−0.008 [−0.056, +0.051]	333	< 0.001	0.638	1
Vertical velocity (v, m s^{-1})	POD-GPR	6.48	+0.026 [−0.035, +0.079]	333	< 0.001	0.665	1
Vertical velocity (v, m s^{-1})	Domain-informed Concat-MLP	6.95	+0.035 [+0.018, +0.066]	333	< 0.001	< 0.001	0.00207
Vertical velocity (v, m s^{-1})	PINN data-only	7.1	+0.038 [+0.023, +0.066]	333	< 0.001	< 0.001	< 0.001
Vertical velocity (v, m s^{-1})	Random Forest	7.62	+0.046 [+0.015, +0.078]	333	< 0.001	0.0715	0.286
Vertical velocity (v, m s^{-1})	XGBoost	7.86	+0.044 [−0.005, +0.116]	333	< 0.001	0.0315	0.157
Vertical velocity (v, m s^{-1})	Latent Neural Operator	8.17	+0.066 [+0.040, +0.084]	333	< 0.001	< 0.001	< 0.001
Vertical velocity (v, m s^{-1})	Nearest case	8.19	+0.026 [−0.038, +0.109]	333	< 0.001	0.214	0.641
Vertical velocity (v, m s^{-1})	DeepONet	8.93	+0.086 [+0.046, +0.107]	333	< 0.001	< 0.001	0.00282
Vertical velocity (v, m s^{-1})	CatBoost	9.45	+0.108 [+0.019, +0.138]	333	< 0.001	0.00865	0.0519
Vertical velocity (v, m s^{-1})	Extra Trees	10	+0.103 [+0.076, +0.153]	333	< 0.001	< 0.001	< 0.001
Vertical velocity (v, m s^{-1})	PINN continuity	10	+0.100 [+0.067, +0.148]	333	< 0.001	< 0.001	< 0.001
Vertical velocity (v, m s^{-1})	GAMS-GNN-style	12.1	+0.123 [+0.092, +0.168]	333	< 0.001	< 0.001	< 0.001
Vertical velocity (v, m s^{-1})	Transolver-style	12.2	+0.114 [+0.091, +0.178]	333	< 0.001	< 0.001	< 0.001
Vertical velocity (v, m s^{-1})	U-FNO3D	14.6	+0.184 [+0.148, +0.228]	333	< 0.001	< 0.001	< 0.001
Vertical velocity (v, m s^{-1})	F-FNO	15.3	+0.217 [+0.179, +0.266]	333	< 0.001	< 0.001	< 0.001
Vertical velocity (v, m s^{-1})	Ridge	17	+0.598 [+0.471, +0.673]	333	< 0.001	< 0.001	< 0.001
Vertical velocity (v, m s^{-1})	Mean baseline	17.3	+0.577 [+0.501, +0.686]	333	< 0.001	< 0.001	< 0.001
Transverse velocity (w, m s^{-1})	POD-RBF	2.9	−0.097 [−0.136, −0.068]	429	< 0.001	< 0.001	< 0.001
Transverse velocity (w, m s^{-1})	POD-GPR	3.67	−0.087 [−0.124, −0.069]	429	< 0.001	< 0.001	< 0.001
Transverse velocity (w, m s^{-1})	Nearest case	4.24	−0.076 [−0.093, −0.064]	429	< 0.001	< 0.001	< 0.001
Transverse velocity (w, m s^{-1})	XGBoost	6.29	+0.005 [−0.002, +0.018]	429	< 0.001	0.457	0.457
Transverse velocity (w, m s^{-1})	Random Forest	6.98	+0.029 [+0.010, +0.045]	429	< 0.001	0.138	0.276
Transverse velocity (w, m s^{-1})	Domain-informed FiLM-MLP	7.02	– (FiLM ref.)	429	< 0.001	– (FiLM ref.)	– (FiLM ref.)
Transverse velocity (w, m s^{-1})	Extra Trees	7.71	+0.033 [+0.018, +0.054]	429	< 0.001	0.0657	0.197
Transverse velocity (w, m s^{-1})	CatBoost	9.26	+0.035 [+0.019, +0.048]	429	< 0.001	0.0164	0.0957
Transverse velocity (w, m s^{-1})	Mean baseline	9.83	+0.048 [+0.030, +0.064]	429	< 0.001	0.0176	0.0957
Transverse velocity (w, m s^{-1})	Domain-informed Concat-MLP	9.83	+0.041 [+0.024, +0.051]	429	< 0.001	< 0.001	< 0.001
Transverse velocity (w, m s^{-1})	PINN data-only	10.6	+0.038 [+0.025, +0.051]	429	< 0.001	< 0.001	< 0.001

Target field	Model	Mean rank	Median Δ vs. FiLM [95% CI]	Friedman χ^2_F	Friedman p	Wilcoxon raw p	Holm p
Transverse velocity (w, m s^{-1})	PINN continuity	11.1	+0.049 [+0.032, +0.065]	429	< 0.001	0.0137	0.0957
Transverse velocity (w, m s^{-1})	Ridge	11.5	+0.051 [+0.031, +0.066]	429	< 0.001	0.0147	0.0957
Transverse velocity (w, m s^{-1})	Latent Neural Operator	13.2	+0.068 [+0.048, +0.073]	429	< 0.001	< 0.001	< 0.001
Transverse velocity (w, m s^{-1})	DeepONet	13.4	+0.054 [+0.038, +0.079]	429	< 0.001	< 0.001	< 0.001
Transverse velocity (w, m s^{-1})	GAMS-GNN-style	13.8	+0.067 [+0.049, +0.076]	429	< 0.001	0.00175	0.014
Transverse velocity (w, m s^{-1})	Transolver-style	14.5	+0.067 [+0.057, +0.081]	429	< 0.001	0.00132	0.0118
Transverse velocity (w, m s^{-1})	U-FNO3D	16.8	+0.086 [+0.074, +0.109]	429	< 0.001	< 0.001	< 0.001
Transverse velocity (w, m s^{-1})	F-FNO	17.5	+0.101 [+0.086, +0.116]	429	< 0.001	< 0.001	< 0.001
Gauge pressure (p_{gauge}, kPa)	POD-RBF	4.83	−10.116 [−25.553, −6.378]	180	< 0.001	0.0196	0.255
Gauge pressure (p_{gauge}, kPa)	Extra Trees	6.1	−7.507 [−20.564, −2.842]	180	< 0.001	0.204	1
Gauge pressure (p_{gauge}, kPa)	XGBoost	6.98	−8.154 [−24.205, −5.315]	180	< 0.001	0.321	1
Gauge pressure (p_{gauge}, kPa)	POD-GPR	7.69	−5.706 [−10.477, +10.575]	180	< 0.001	0.795	1
Gauge pressure (p_{gauge}, kPa)	Random Forest	8.14	−8.229 [−21.200, −4.340]	180	< 0.001	0.536	1
Gauge pressure (p_{gauge}, kPa)	CatBoost	8.36	−3.560 [−22.427, +20.593]	180	< 0.001	0.951	1
Gauge pressure (p_{gauge}, kPa)	Domain-informed FiLM-MLP	8.79	– (FiLM ref.)	180	< 0.001	– (FiLM ref.)	– (FiLM ref.)
Gauge pressure (p_{gauge}, kPa)	Latent Neural Operator	9.69	+2.713 [−1.261, +8.239]	180	< 0.001	0.223	1
Gauge pressure (p_{gauge}, kPa)	Transolver-style	10.1	+1.406 [−1.399, +8.926]	180	< 0.001	0.228	1
Gauge pressure (p_{gauge}, kPa)	GAMS-GNN-style	10.2	+3.719 [−4.746, +8.037]	180	< 0.001	0.258	1
Gauge pressure (p_{gauge}, kPa)	PINN continuity	10.2	+6.587 [+1.216, +19.627]	180	< 0.001	0.00304	0.0426
Gauge pressure (p_{gauge}, kPa)	Nearest case	10.3	−3.968 [−8.065, +38.029]	180	< 0.001	0.148	1
Gauge pressure (p_{gauge}, kPa)	PINN data-only	10.4	+4.533 [−1.913, +11.145]	180	< 0.001	0.346	1
Gauge pressure (p_{gauge}, kPa)	Domain-informed Concat-MLP	10.7	+3.667 [+1.911, +7.764]	180	< 0.001	0.28	1
Gauge pressure (p_{gauge}, kPa)	DeepONet	10.8	+6.131 [+2.596, +14.928]	180	< 0.001	0.0757	0.908
Gauge pressure (p_{gauge}, kPa)	U-FNO3D	13.8	+19.108 [+12.085, +51.431]	180	< 0.001	< 0.001	0.00513
Gauge pressure (p_{gauge}, kPa)	Mean baseline	14.1	+134.816 [+44.495, +253.078]	180	< 0.001	< 0.001	< 0.001
Gauge pressure (p_{gauge}, kPa)	F-FNO	14.3	+30.349 [+20.567, +51.663]	180	< 0.001	< 0.001	0.00489
Gauge pressure (p_{gauge}, kPa)	Ridge	14.5	+41.582 [+28.904, +69.324]	180	< 0.001	< 0.001	0.00103